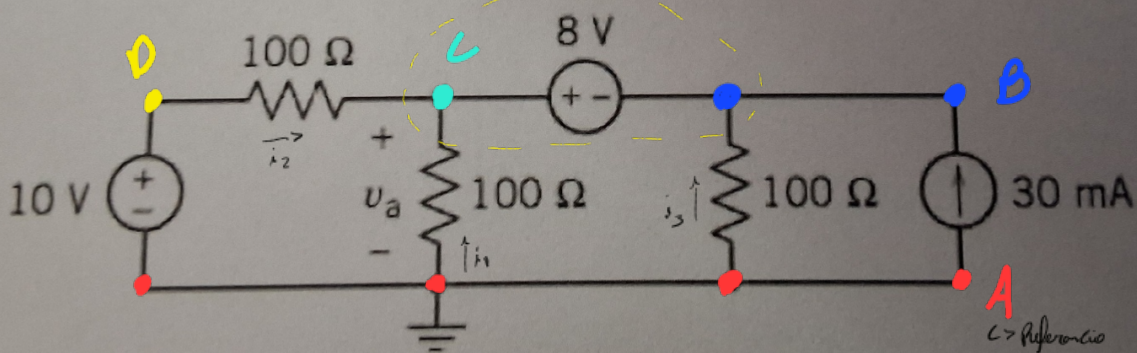


Resposta: $v_a = 7 \text{ V}$

$$V = R \cdot i$$



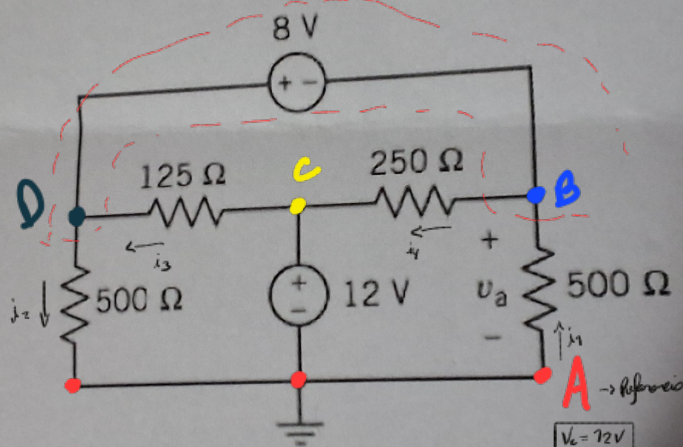
$$V_D = 10 \text{ V} \quad i_1 = \frac{-V_C}{100} \quad i_2 = \frac{10 - V_C}{100} \quad i_3 = \frac{-V_B}{100} \quad V_C - V_B = 8 \text{ V} \rightarrow -V_B = 8 - V_C \rightarrow V_B = V_C - 8$$

$$\text{* Superando CB: } i_1 + i_2 + i_3 + 0,03 = 0 \rightarrow \frac{-V_C}{100} + \frac{10 - V_C}{100} - \frac{(V_C - 8)}{100} + 0,03 = 0 \quad (\times 100) \rightarrow -V_C + 10 - V_C - V_C + 8 + 3 = 0 \rightarrow -3V_C = -21 \rightarrow V_C = 7 \text{ V} \therefore V_a = 7 \text{ V}$$

Resposta: $v_a = 4 \text{ V}$

$$V_D - V_B = 8$$

$$V_D = 8 + V_B$$



$$i_1 = \frac{-V_B}{500} \quad i_2 = \frac{V_D}{500} \quad i_3 = \frac{12 - V_D}{125} \quad i_4 = \frac{V_B - 12}{250}$$

$$\text{* Superando BD: } i_3 + i_1 = i_4 + i_2$$

$$\frac{12}{125} - \frac{(8 + V_B)}{125} - \frac{V_B}{500} = \frac{V_B}{250} - \frac{12}{250} + \frac{8}{500} + \frac{V_B}{500} \quad (\times 500)$$

$$48 - 32 - 4V_B - V_B = 2V_B - 24 + 8 + V_B$$

$$16 - 5V_B = 3V_B - 16$$

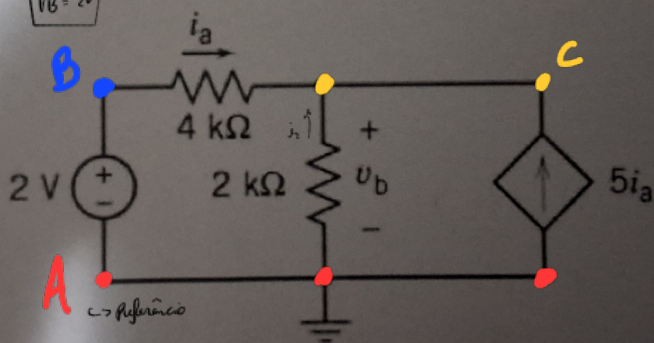
$$-8V_B = -32$$

$$V_B = 4 \text{ V}$$

$$\text{* Como A é referência } V_B = V_D = 4 \text{ V}$$

Resposta: $v_b = 1,5 \text{ V}$

$$V_B = 2 \text{ V}$$



$$i_1 = \frac{2 - V_C}{2000}$$

$$\text{* Lei de KCL: } 6i_1 + i_2 = 0$$

$$i_2 = \frac{-V_C}{4000}$$

$$6 \left(\frac{2 - V_C}{2000} \right) - \frac{V_C}{4000} = 0 \quad (\times 4000)$$

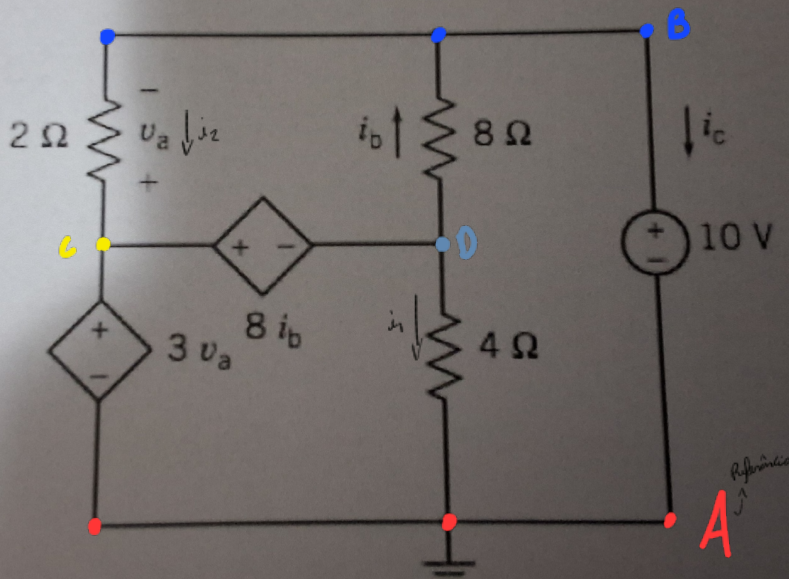
$$12 - 6V_C - 2V_C = 0$$

$$-8V_C = -12$$

$$V_C = \frac{12}{8} = 1,5 \text{ V}$$

$$\therefore V_b = V_C = 1,5 \text{ V}$$

P 4.4-7 Determine o valor da corrente i_c na Figura P 4.4-7.



$$V_B = 10V$$

$$i_b = \frac{V_D - 10}{8} \quad i_1 = \frac{V_D}{4} \quad i_2 = \frac{10 - V_C}{2}$$

$$V_D = V_C - 10$$

$$V_C = 3(V_C - 10)$$

$$V_C = 7.5V$$

$$V_C - V_D = 8 \left(\frac{V_D - 10}{8} \right)$$

$$7.5 - V_D = V_C - 10$$

$$7.5 = 2V_D$$

$$V_D = 3.75V$$

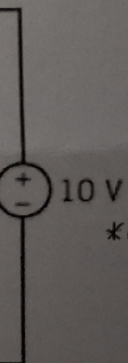
$$i_c + i_2 = i_b$$

$$i_c = i_b - i_2$$

$$i_c = \left(\frac{7.5 - 10}{8} \right) - \left(\frac{10 - 7.5}{2} \right)$$

$$i_c = -2.875A$$

ecida pela fonte de-



* corrente que sai do nó C $i_1 + i_2 \rightarrow i_g = \frac{5.455 - 10}{4} + \frac{5.455}{12} = -0.687A$

$P = (-0.687)(-1.515) = 1.031W$
 $\downarrow$
 $V_C - V_D$

$$V_B = 10$$

$$i_D = \frac{V_D - 10}{16} \quad i_1 = \frac{V_C - 10}{4} \quad i_2 = \frac{V_D}{8} \quad i_3 = \frac{V_C}{12}$$

$$V_C - V_D = 8 \cdot i_D$$

$$V_C - V_D = \frac{V_D}{2} - 5 \quad (x2)$$

$$\rightarrow 2V_C - 2V_D = V_D - 10$$

$$2V_C = 3V_D - 10$$

$$V_C = \frac{3}{2}V_D - 5 = 5.455$$

* Superlucro CD: $i_1 + i_D + i_2 + i_3 = 0$

$$\left(\frac{3}{2}V_D - 5 \right) + \frac{V_D - 10}{16} + \frac{V_D}{8} + \frac{3}{12}V_D - 5 = 0$$

$$\frac{3V_D - 10}{8} + \frac{V_D - 10}{16} + \frac{V_D}{8} + \frac{3V_D - 10}{24} = 0 \quad (x24)$$

$$9V_D - 10 + \frac{3V_D}{2} - 15 + 3V_D + 3V_D - 10 = 0$$

$$15V_D + \frac{3V_D}{2} - 115 = 0 \xrightarrow{x2} 33V_D - 230 = 0 \rightarrow V_D = 6.97V$$